

HISTORICAL

Air Quality Analysis

9,357 hourly observations · 13 months · Air quality & meteorology

Trends · Relationships · Predictive insight

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UCI Machine Learning Repository · De Vito et al.

AirQualityUCI · Italian city · Mar 2004 – Apr 2005

Five things this dataset tells us

Headline findings before we go deep

01

Pronounced diurnal pattern

Twin pollution peaks occur around 08:00 and 19:00. Mean CO at the evening peak is $\sim 5\times$ the 05:00 trough.

02

Benzene is elevated

Mean benzene is $\sim 10 \mu\text{g}/\text{m}^3$, approximately $2\times$ the EU calendar-year limit value of $5 \mu\text{g}/\text{m}^3$.

03

Strong pollutant cluster

$\text{CO} \leftrightarrow \text{benzene } r = 0.91$ and $\text{CO} \leftrightarrow \text{NO}_x r = 0.80$, revealing a strong combustion-related pollutant cluster.

04

Weekend NO_x falls 27 %

A pronounced weekly pollution pattern consistent with reduced weekend traffic activity.

05

Benzene is predictable

Random Forest achieves $R^2 \approx 0.77$ using CO + NO_x + NO₂ + weather + temporal features.

The dataset

What we received and how we cleaned it

9,357

hourly
observations

389

days
covered

13

raw
variables

~9 %

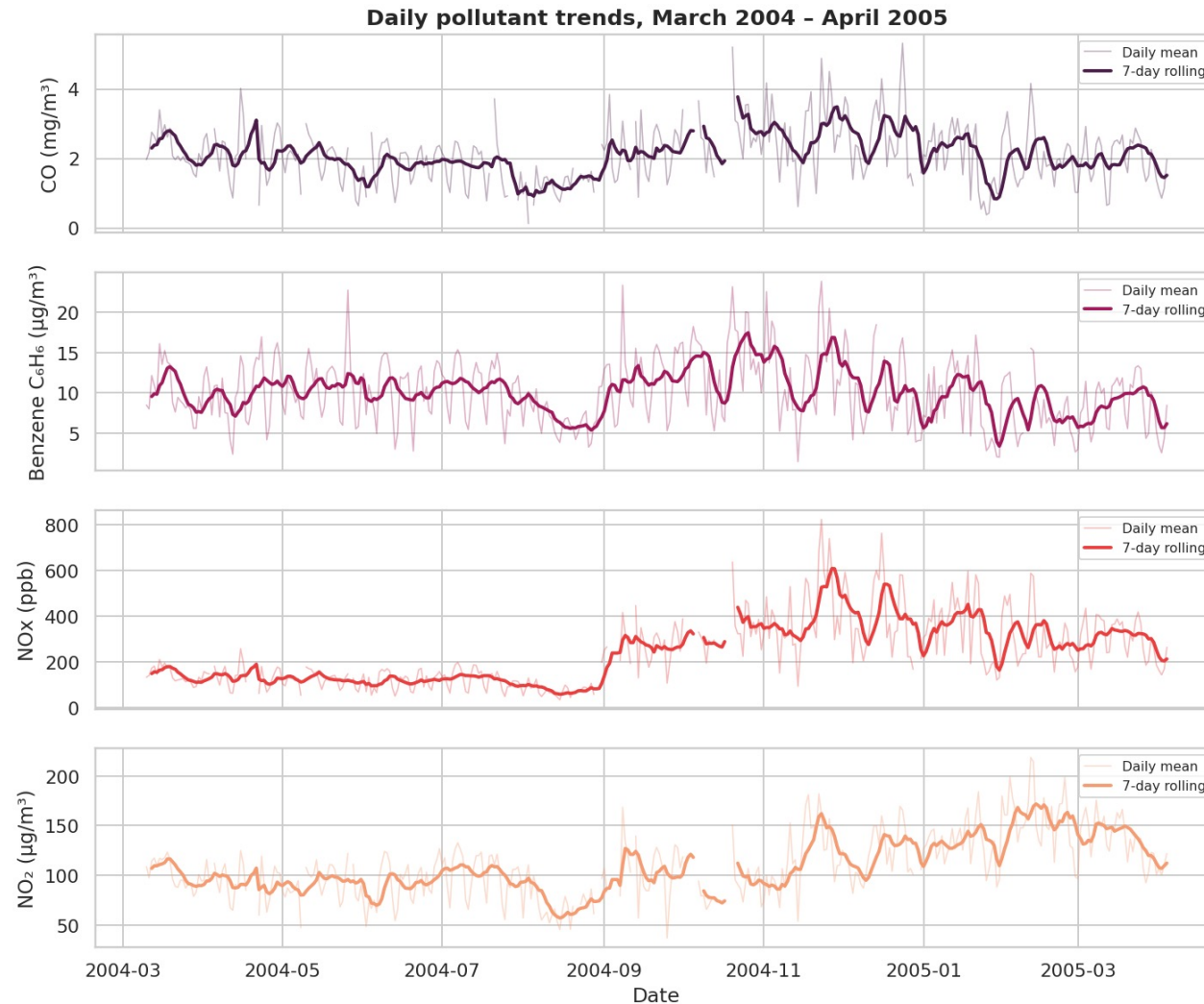
missing after
cleaning

Cleaning steps

- Replaced the -200 sentinel with NaN throughout
- Combined Date + Time into a Python datetime index
- Dropped NMHC (>90 % missing)
- Time-based interpolation applied with a six-observation limit in each direction; unresolved missing values retained.
- Removed two stray empty columns and 114 blank trailing rows

Daily trends across 13 months

Winter peaks for NO_x and NO₂; summer dip for CO and benzene



WHAT TO NOTICE

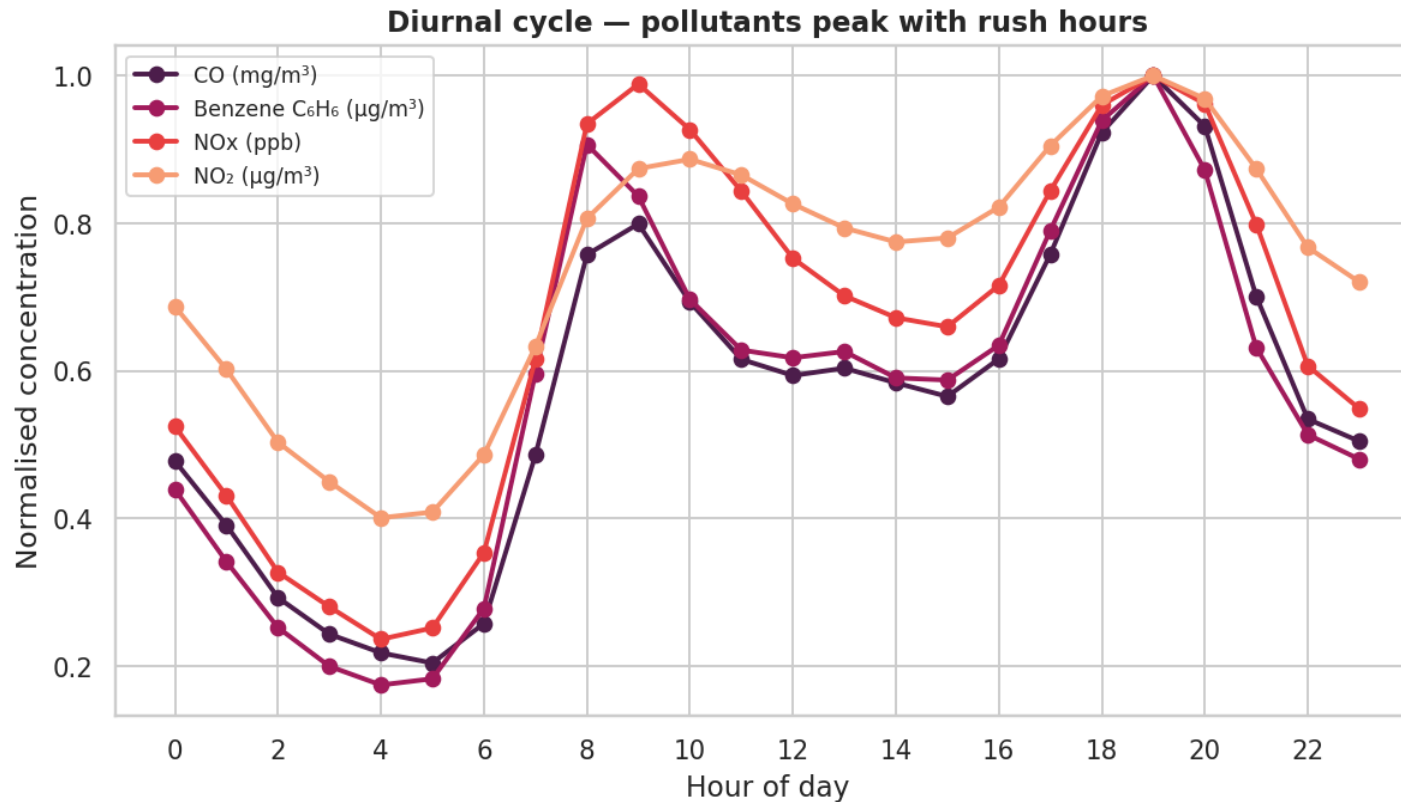
NO_x and NO₂ climb sharply from October, consistent with seasonal changes in emissions and atmospheric dispersion.

CO and benzene dip in late summer.

The dataset ends *before the 2005 summer trough*, limiting inter-annual analysis.

The rush-hour fingerprint

Twin peaks at 08:00 and 19:00 reveal a pronounced diurnal pollution pattern



CO MEAN

3.72 mg/m³

at 19:00 — evening peak

0.76 mg/m³

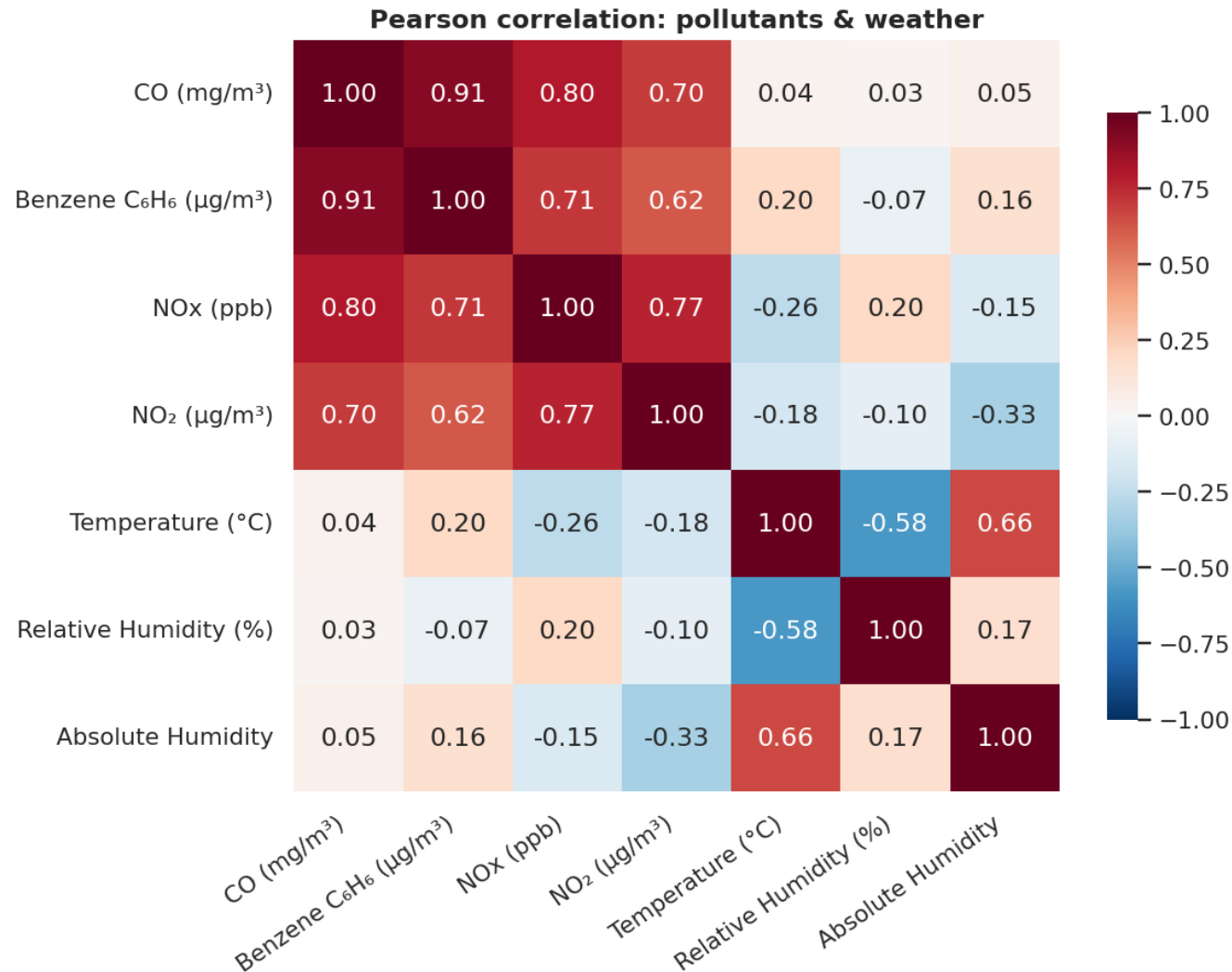
at 05:00 — overnight trough

≈ 5× difference between
rush hour and dawn

NO₂ remains elevated through the afternoon, giving it a broader daytime profile than the other pollutants.

A strong combustion-related pollutant cluster

The combustion cluster shows up cleanly in the correlation matrix



Combustion cluster

CO · Benzene · NOx · NO₂

$r \approx 0.6-0.9$

Weather block

T ↔ AH positive, T ↔ RH negative

physics, not pollution

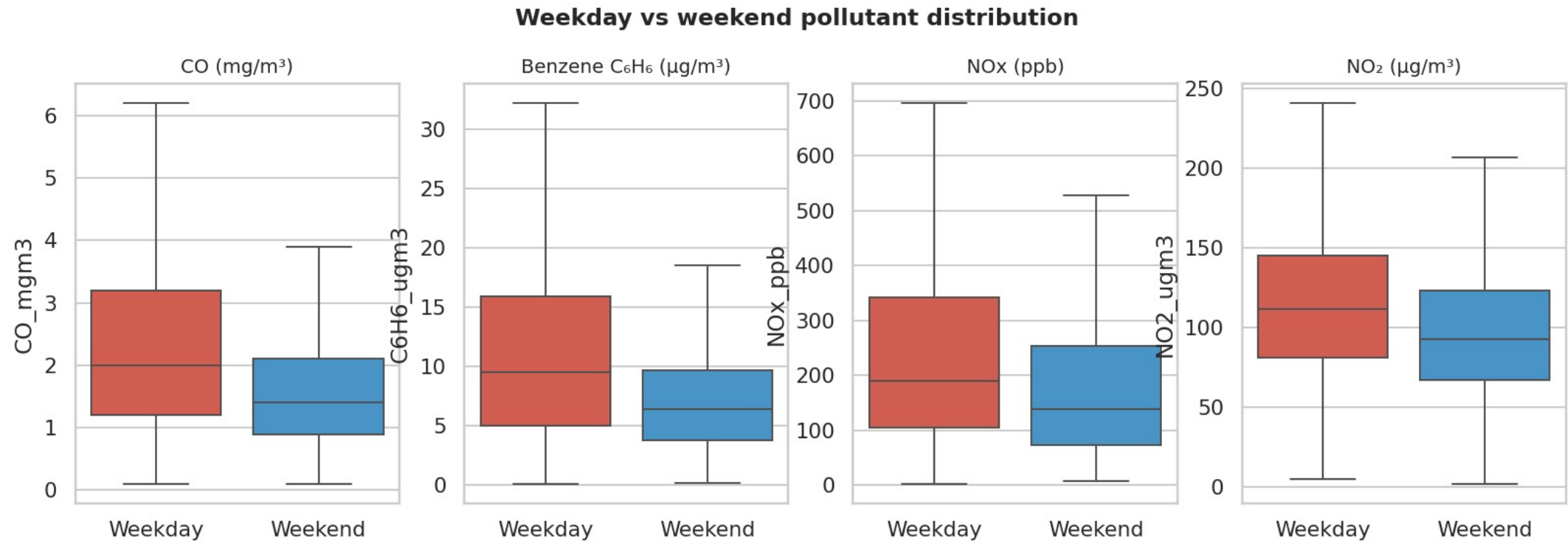
Cross terms

NOx & NO₂ vs Temperature

weak negative association

Weekend Nox is substantially lower

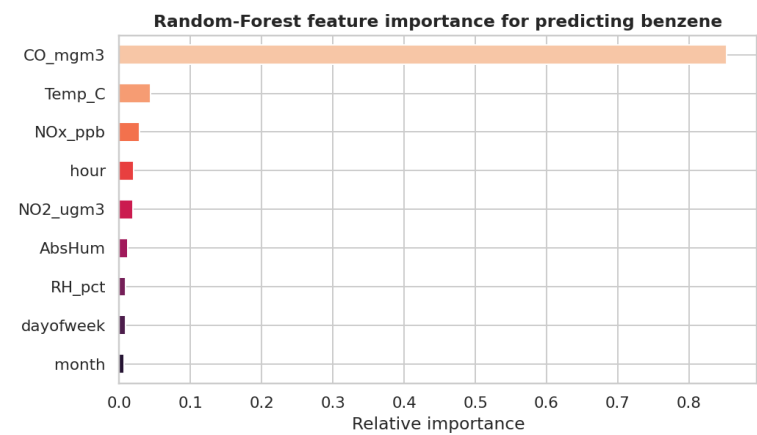
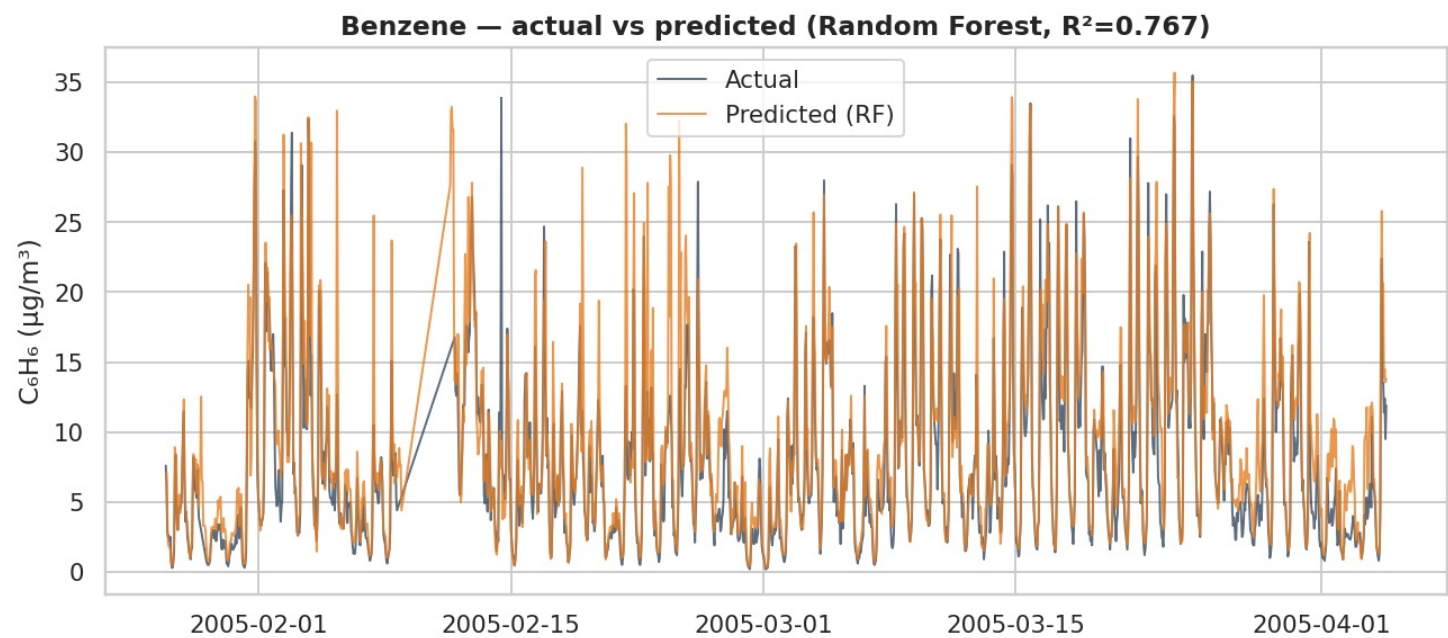
A clear weekly shift in the pollutant distribution



-27 % median weekday-to-weekend NOx drop, *highlighting the potential importance of traffic activity in urban pollution patterns.*

Predictive Modelling

If a station has CO + NOx + NO₂ + weather, it can estimate benzene with useful accuracy



MODEL PERFORMANCE

Linear Regression

R²

0.77

MAE (µg/m³)

2.26

Random Forest

R²

0.77

MAE (µg/m³)

1.76

CO is the dominant predictor, consistent with its strong correlation with benzene ($r = 0.91$).

What this means for health and policy

Translating the data into actionable insights

PUBLIC HEALTH

- **Benzene** — mean concentration $\sim 10 \mu\text{g}/\text{m}^3$, approximately twice the EU calendar-year limit value of $5 \mu\text{g}/\text{m}^3$. WHO states that no safe level of benzene exposure can be recommended. WHO classifies benzene as a Group-1 human carcinogen.
- **NO₂** — hourly concentrations reach $340 \mu\text{g}/\text{m}^3$, with some observations above $200 \mu\text{g}/\text{m}^3$. Formal exceedance analysis is needed to assess regulatory significance.
- **Vulnerable groups** — children, the elderly, and people with cardiopulmonary disease bear most of the risk.

POLICY LEVERS

- **Traffic activity** — the 27% weekend NO_x reduction provides a strong rationale for investigating traffic-management interventions.
- **Seasonality** — winter NO_x and NO₂ peaks highlight the importance of seasonal emissions and atmospheric dispersion.
- **Monitoring** — benzene can be estimated from CO + NO_x + NO₂ + weather ($R^2 \approx 0.77$), highlighting the potential for denser, lower-cost monitoring networks.

Limitations and next steps

Honest framing of what the data can and cannot tell us

L I M I T A T I O N S

- **Single site** — results don't generalise across cities.
- **13 months of coverage** — too short for inter-annual trend or policy-impact analysis.
- **NMHC dropped** — >90 % missing leaves total non-methane hydrocarbons unmeasured.
- **Confounded weather/season** — traffic patterns and meteorology co-vary seasonally.

N E X T S T E P S

- **Validate** against contemporaneous regulatory monitoring stations.
- **Extend the model** to ozone (O₃) using the PT08.S5 sensor.
- **Formal trend analysis** — seasonal ARIMA or state-space decomposition.
- **NO₂ exceedance analysis** — assess hourly concentrations using the applicable regulatory threshold, averaging period and permitted exceedance frequency.

Thank you

Questions and discussion

DELIVERABLES

Cleaned dataset · Daily / monthly aggregations · Reproducible Jupyter notebook · Full report with eight figures

DATA & TOOLS

UCI Air Quality dataset · Python · pandas · matplotlib · seaborn · scikit-learn

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